

Cara Kaissar . Jarvis Consulting

As a passionate software engineer with a strong academic background and professional experience, I hold a bachelor's degree in Computer Science and Software Engineering from UQAM. Currently, I am part of Jarvis' Technical Consulting (TIC) Program and have a solid foundation in both backend and front-end development, thanks to my experience with Astuce Media. My journey in the software industry is driven by a keen interest in innovative technologies and a desire to create efficient, scalable solutions. I am particularly passionate about leveraging technology to solve real-world problems and enhance user experiences, with a special enthusiasm for machine learning and data-driven solutions.

Skills

Proficient: Java, Linux/Bash, RDBMS/SQL, Agile/Scrum, Git

Competent: Python, JavaScript, HTML/CSS, Docker, RESTful APIs, Data Analysis, Data Modeling, Database Normalization (1NF, 2NF, 3NF), SQL Joins and Subqueries, SQL Aggregation Functions

Familiar: Pandas/Numpy, Scikit-Learn, Machine Learning : Supervised Learning, Unsupervised Learning, Cloud Computing (AWS/GCP), Data Visualization (Matplotlib/Seaborn)

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_demo

Cluster Monitor [GitHub]: Developed a comprehensive cluster monitoring tool using Bash and SQL to manage and analyze system performance metrics. Implemented data collection scripts to capture CPU, memory, and disk usage across multiple Linux servers. Utilized PostgreSQL to store and query collected data, providing real-time insights and historical trends. Created automated alerting mechanisms for critical thresholds, ensuring proactive system maintenance. Leveraged crontab for scheduled data collection and reporting, enhancing the tool's reliability and efficiency.

RDBMS and SQL [GitHub]: Developed and managed relational databases using PostgreSQL. Acquired skills in CRUD operations, data modeling, and normalization/denormalization principles. Designed complex SQL queries for data manipulation and extraction. Implemented joins, subqueries, and aggregations to handle data relationships and perform advanced analysis. Learned key concepts such as entity relationships, primary and foreign keys, and normalization forms (1NF, 2NF, 3NF). Utilized tools and resources such as pgeercises and LinkedIn Learning for practical exercises and theoretical understanding.

Core Java Apps [GitHub]:

- **Grep App:** Implemented a Java-based grep application for text pattern searching within files. The project involved creating a basic grep functionality and then re-implementing it using Java 8 Lambda and Stream APIs for better memory efficiency and performance. The app is capable of processing large files efficiently and was containerized using Docker for consistent deployment across various environments.
- **JDBC App:** Developed a Java application that retrieves real-time stock data using an external API, processes the information, and stores it in a local database using JDBC. The project involved integrating a REST API to fetch stock data, error handling, and data persistence. The app was containerized using Docker for consistent deployment across different environments.

Highlighted Projects

Assistant for Serial Histology of Mouse Brains (SBH-Assistant) [GitHub]: Collaborated with a team of three to develop an assistant software for serial histology of mouse brains using OCT. My primary responsibility was building the API and backend application that manages the database and synchronizes services. This tool aids users in planning serial histology experiments, assists during imaging sessions, and documents experiments. It integrates the Allen Mouse Brain atlas and supports 2D-3D image registration, graphical visualization of reference data, regions of interest, and slices added during the experiments.

Professional Experiences

Technical Consultant - Developer Role, Jarvis (June 2024-present): Designed and implemented robust software solutions, focusing on backend development using Java and SQL. Collaborated with cross-functional teams to develop and deploy scalable applications. Utilized Agile methodologies, specifically Scrum, to ensure timely delivery of features and

enhancements. Developed a cluster monitoring tool to track and analyze system performance metrics, employing Bash scripting and PostgreSQL. Leveraged Gitflow for efficient version control and collaborative development.

Junior Software Engineer (Promoted From Intern), AstuceMedia (May 2023 - May 2024): Developed web applications using React and Node.js. Built and optimized backend services with Node.js and Express. Implemented RESTful APIs for frontend-backend communication. Managed data using MongoDB and MySQL. Integrated front-end and back-end technologies effectively. Applied design patterns like Singleton and Observer in Angular. Integrated REST APIs with Angular applications. Collaborated with teams to design and launch features. Conducted performance testing and debugging.

Education

UNIVERSITÉ DU QUÉBEC À MONTRÉAL (UQAM) (2019 - 2024), Bachelor of Science in Computer Science and Software Engineering, Computer Science and Software Engineering

ÉCOLE FRANÇAISE INTERNATIONALE DE RIYADH (2002 - 2017), French Baccalauréat, Série S (Scientifique), Science

Miscellaneous

- Basketball
- Tennis
- Gym / Pilates / Yoga
- Spending Quality Time With My Family & Friends
- History Books / Documentaries